

Jonas Lim Hong Xiang

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Python software engineer with 7 years designing and delivering production-grade systems end-to-end – from requirements and architecture through deployment, monitoring, and iteration. Proficient in Python (FastAPI), SQL, data analysis pipelines, and ML/AI integration on AWS. Experienced partnering directly with business and product stakeholders to translate complex requirements into robust, analytically rigorous software solutions.

PROFESSIONAL EXPERIENCE

Co-Founder & Software Architect

May 2024 - Jun 2026

NockHub, Singapore | nockhub.com

- Designed and built production Python backend services using FastAPI and AWS serverless infrastructure (Lambda, API Gateway, DynamoDB), delivering 40+ data schemas and 30+ REST APIs.
- Collaborated with business owners and non-technical stakeholders to gather requirements, define priorities, and translate them into software specifications and system architecture.
- Led peer code reviews and engineering delivery standards across a team of 3 engineers; oversaw system integration testing and production support across all services.
- Integrated GenAI/LLM automation workflows to eliminate manual repetitive processes and improve operational efficiency.
- Maintained production observability, cost controls, and deployment pipelines across all AWS workstreams.
- Grew platform to 4 ranges, 7,000+ users, and 4,000+ bookings per month.

Senior Machine Learning Engineer

Jun 2022 - Jul 2024

Carro, Singapore

- Delivered production Python systems for data-driven decisioning across pricing analytics, anomaly detection, and operational intelligence – serving 12,000+ requests per day on AWS.
- Led peer code reviews and provided technical mentoring across the ML engineering team; championed engineering best practices and continuous improvement.
- Designed and maintained SQL and PySpark data analysis pipelines, a feature store, and a metadata store; improved data quality by 50% and enabled reproducible analytical workflows.
- Established evaluation frameworks, automated deployment workflows, and monitoring practices; introduced drift detection and reduced operational overhead by 75 engineer-hours per month.
- Replaced monolithic data crawler with distributed architecture; reduced pipeline processing from 7 days to 9 hours with improved data lineage and reliability.
- Optimized AWS infrastructure (networking, compute, databases); delivered USD 120k in cost savings in 2023.
- Partnered with product managers, operations, and data teams across Singapore, Malaysia, Indonesia, and Taiwan to align technical solutions with business objectives.

Data Scientist

Jun 2019 - May 2022

Micron Technology, Singapore

- Built and deployed predictive quantitative models for yield optimization and defect classification within a regulated manufacturing environment; contributed to USD 11M in savings.
- Built SQL analytics pipelines for model training and operational reporting; reduced query time by 50% and improved data reliability.
- Delivered end-to-end software solutions from requirements gathering through production deployment, collaborating with engineering, quality, and operations teams.

EDUCATION & AWARDS

Master of Computer Science (Data Science & AI)

Aug 2024 - Jun 2026

University of Sydney, Australia
Distinction | Singapore IMDA SG Digital Scholar

- Capstone: Designed and deployed an Agentic AI system on a Unitree Go2 quadruped robot – built LLM-driven multi-agent workflows with tool use, task decomposition, and autonomous real-world execution.

Bachelor of Engineering (Aerospace)

Aug 2015 - May 2019

Nanyang Technological University, Singapore
First Class Honors, Highest Distinction

TECHNICAL EXPERTISE

Languages: Python, SQL, TypeScript, Dart, Bash
ML/AI: PyTorch, TensorFlow, Hugging Face, GenAI, Agentic AI
Frontend / Mobile: React, Flutter
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, RDS

Python Frameworks: FastAPI, Pandas, PySpark, Flask
Cloud & Infrastructure: AWS, GCP, Terraform, Docker
Data Engineering: ETL pipelines, feature stores, data quality & lineage
Practices: Agile delivery, CI/CD, code review, production monitoring

HACKATHONS

2025 Global Health Systems Hackathon Hub, "Building High-Value Health Systems: Leveraging AI". Team I Know You Know won 2nd place in Sydney and placed Top 20 globally.

CERTIFICATIONS

AWS Certified Solutions Architect - Professional

Jan 2024

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